# Complex Numbers, Gravity, and the Proposed Hidden Sector

**Gary Abraham Bernstein**

Independent Researcher | ORCID: 0009-0009-1761-2867

## Abstract

Quantum physics uses numbers with real and imaginary components. Could an unseen imaginary part explain gravity or dark matter? The notation alone does not supply that explanation: a change of mathematical phase can mix the two components without changing observations. A physical split therefore needs a rule that gives the components different roles. This paper tests that proposal with simple models and separates it from questions about quantum gravity, singularities and how systems combine. Complex numbers can provide useful structure; the model must show which observations that structure explains.

## 1. What the imaginary unit establishes

Quantum theory uses complex numbers: numbers with real and imaginary components. They describe how contributions combine and interfere. Calling a component imaginary does not mean it is an unseen material. The question here is whether a physical model can turn that mathematical distinction into an explanation of gravity or dark matter.

There are two different questions. Can a complex quantity be written as a pair of real quantities? Algebraically, yes. Does a particular theory on real Hilbert spaces reproduce the same composition, preparation and measurement structure as standard complex quantum theory? That question requires analysis of the complete theory.

Calling complex numbers mere notation can miss a physically consequential structure. Calling their use proof of a hidden imaginary mass sector goes beyond the same evidence. The proposal must identify which transformations and observables have physical significance before assigning separate ontological roles to the two coordinate components.

The four research areas considered here can inform one another. They should not be treated as four independent confirmations of one mechanism until that mechanism has been defined. A shared mathematical vocabulary is the beginning of a possible unification, not its result.

## 2. Complex variables in gravitational theory

General relativity is conventionally formulated with a real Lorentzian metric. Quantum states are conventionally represented in complex Hilbert spaces. This

contrast can motivate a search for common structures, but it is not a proof that their physical incompatibilities reduce to choosing different number systems.

Twistor theory supplies a complex-geometric reformulation of aspects of spacetime and field theory (Penrose, 1967). Complex connection variables in Ashtekar's formulation can simplify the gravitational constraints, while reality conditions are needed to recover the intended real Lorentzian physics (Ashtekar, 1986). Those conditions are part of specifying the physical theory; their presence cannot simply be dismissed as an unjustified restriction on reality.

Complex metrics also enter gravitational path-integral questions. Witten's discussion of allowable complex spacetime metrics concerns which complex configurations can be admitted under specified conditions (Witten, 2021). It does not establish that every complex metric is a physical spacetime or that complexification alone completes quantum gravity.

These approaches make complex structure worth taking seriously. They do not yet provide a common map between their variables, state spaces, observables and dynamics. A proposed unification must supply that map and show what it predicts. Similar-looking complex notation does not ensure that two theories assign the same physical meaning to it.

Wick rotation illustrates the distinction. Under appropriate analytic assumptions, rotating a time variable can relate Lorentzian and Euclidean calculations. In curved spacetime there need not be a single globally preferred operation of this kind. Its validity and interpretation belong to the particular model, not to the mere existence of the complex plane.

### 2.1 What a common complex geometry would have to unify

Twistor and connection approaches deserve more than being counted as occurrences of the symbol i. Their attraction is that changing variables can expose relations obscured in a metric-only description. A successful change of variables can simplify constraints, reveal symmetries and make previously difficult calculations possible. This is a substantive reason to investigate complex geometry, even when the resulting formulation remains physically equivalent to a real description.

The proposed unification should therefore specify three maps. First, it must recover an observable Lorentzian metric with the intended signature. Second, it must identify the quantum state space, composition rules and observables. Third, it must recover the relevant dynamics, including the interaction between matter and geometry. A formal map at only one level is not a solution to the other two problems. Reality conditions, where required, are part of the first map, not evidence that the physical theory has artificially discarded a dark half of itself.

This leaves room for an ambitious interpretation. The repeated usefulness of

complex structures may indicate that the right fundamental description makes phase and geometry aspects of one construction. But the supporting evidence consists of the relations actually demonstrated by each programme. It cannot be pooled as independent confirmation of an unspecified real-visible/imaginary-dark projection. The stronger the proposed common cause, the more important its explicit mathematical identification becomes.

## 3. The phase test for a hidden imaginary sector

Let a complex field be written as Phi = a + ib, with real components a and b. A phase rotation through theta gives

$$a' = a\cos\theta - b\sin\theta, \qquad b' = a\sin\theta + b\cos\theta.$$

The modulus is invariant: a-prime squared plus b-prime squared equals a squared plus b squared. The separate real and imaginary components are not invariant. A field written as purely imaginary in one phase convention can be written as purely real in another.

If such a phase change is a gauge redundancy, observable darkness cannot depend on which redundant description is selected. If it is a physical global symmetry, a proposal must still state what interaction or symmetry-breaking structure selects one component for a different coupling. A preferred real-axis detector is additional dynamics, not a consequence of complex notation.

For a charged complex scalar, minimal electromagnetic coupling uses a covariant derivative of the whole field. In a convention where D_mu = partial_mu + iq A_mu,

$$|D_\mu\Phi|^2 = (\partial_\mu a - qA_\mu b)^2 + (\partial_\mu b + qA_\mu a)^2.$$

Both components enter the charged-field interaction. If the field carries no electromagnetic charge, both can be electromagnetically dark. Darkness is determined by the couplings and representation, not by the word imaginary.

Gravity couples to the stress-energy derived from the matter action. For suitable scalar models, that includes modulus and derivative combinations. This does not mean that gravity alone detects one component while electromagnetic instruments automatically project onto the other. The proposed split must be derived from a consistent action that identifies the relevant symmetries and observables.

This is the central obstacle to the earlier imaginary-sector interpretation. It is more specific than a request for caution. It gives a test: transform the fields according to the proposed theory and check whether the visible-dark distinction changes under a transformation that should leave observations unchanged.

## 4. Scalar-field dark matter and what it supports

Ultralight scalar-field dark matter is an established class of physical models, with wave effects on astrophysical scales for appropriate masses (Hui et al., 2017). Complex models can carry a conserved charge under a global U(1) symmetry. Real scalar models also exist; wave behavior is not exclusive to a literal imaginary component.

A model requires a field action, mass, interactions, cosmological preparation and a calculation of observables. Solitonic-core behavior in simulations and small-scale power suppression concern those specified models. They do not independently confirm that reality contains a gravitationally visible imaginary axis.

The same distinction applies to observation. A simulation predicting a core is not a recent observation of a core. A small-scale discrepancy with one cold-dark-matter treatment does not automatically favor every wave model. Astrophysical assumptions and observational bounds must be included in the comparison. Lyman-alpha constraints are tests of specified field masses and cosmological modeling, not measurements of an imaginary sector as such.

Proposals based on metric derivatives may eventually be related to scalar-field models, but an equivalence has not been derived. A phase component, a higher derivative of a metric and a virtual mode under a quantization prescription are different constructions. A common explanation would need to map their actions, degrees of freedom and observables, not merely note that each can be described with complex mathematics.

For the present proposal, the constructive next step is a toy action in which a well-defined physical mechanism separates visible and dark degrees of freedom. It must pass the phase test, conserve the required quantities and produce a calculable gravitational response. Without that action, 'perpendicular matter' remains an image rather than an explanation.

### 4.1 A minimal invariant model: what succeeds and what does not

An illustrative way to make the proposal concrete is to take a complex scalar Phi neutral under electromagnetism. In units with hbar = c = 1 and metric signature (−,+,+,+), use

$$S_\Phi = \int d^4x\sqrt{-g}\left[-g^{\mu\nu}\partial_\mu\Phi^*\partial_\nu\Phi - m^2|\Phi|^2 - \frac{\lambda}{2}|\Phi|^4\right].$$

Here m^2 > 0 and lambda  0 give a simple stable classical potential. A global U(1) phase symmetry is present. Both real components are electromagnetically neutral; this is a model assumption about the representation, not a consequence of their names. Varying the metric yields

$$T_{\mu\nu} = \partial_\mu \Phi^* \partial_\nu \Phi + \partial_\nu \Phi^* \partial_\mu \Phi - g_{\mu\nu} \left( g^{\alpha\beta} \partial_\alpha \Phi^* \partial_\beta \Phi + V \right).$$

This stress-energy is phase invariant and can source gravity. For a homogeneous field in a spatially flat cosmological background, its density and pressure are rho = |dot Phi|^2 + V and p = |dot Phi|^2 − V. In the quadratic, rapidly oscillating regime, with m much larger than the expansion rate and self-interactions negligible, the averaged kinetic and potential contributions balance and the mean pressure approaches zero. That is the elementary route by which a scalar can behave like cold matter at the background level. It is not a calculation of the cosmological abundance or a proof of acceptable perturbations.

The example establishes a consistent limited result: a complex field can gravitate while being electromagnetically dark. It does not establish that its imaginary part alone is dark. In fact, the invariant action makes precisely the opposite point: the darkness belongs to the whole neutral field. A visible charged field would be an additional field with its own covariant derivative and interactions.

Interactions between the sectors can be written, for example a Higgs-portal term proportional to |Phi|^2 H-dagger H, where H is the Standard Model Higgs doublet. Such a term is consistent with the phase symmetry and introduces another coupling. Whether it is sufficiently small for observational bounds, and whether it creates or depletes the required abundance, must be calculated. Excluding it is also a model choice to justify; neutrality alone does not forbid all nongravitational interactions.

This is not presented as a new scalar dark-matter theory. It is a worked consistency example that replaces the unsupported detector-projection mechanism. Its parameters and preparation history are visible on the page, so the interpretation can be tested against an actual action rather than protected by an undefined hidden sector.

### 4.2 How a physically preferred component could arise

A different construction could distinguish two real components through additional structure. Write Phi = (a + ib)/sqrt(2) and add a term Delta V = epsilon(Phi^2 + Phi*^2)/2, with real epsilon. The quadratic potential becomes

$$V_2 = \frac{1}{2}(m^2 + \epsilon)a^2 + \frac{1}{2}(m^2 - \epsilon)b^2.$$

For |epsilon| < m^2 both squared masses are positive. The continuous U(1) symmetry has been broken to a discrete symmetry and the two mass eigenmodes now have different dynamical roles. Renaming the field coordinates cannot erase the physical mass splitting, because the interaction coefficients must transform with the coordinates. This shows what it means to earn a component distinction.

It still does not assign electromagnetic visibility to one mode. That would require specified gauge representations, mixing or further interactions. The example supplies a possible mathematical ingredient, not a completed visible-dark unification. In particular, the same construction cannot simply be added to an electromagnetically charged scalar: Phi^2 would generally violate its gauge symmetry without additional charged structure. Maintaining consistency is more demanding than selecting a real axis on a diagram.

### 4.3 Which astronomical claims follow

Solitonic cores, wave interference and small-scale suppression arise for particular field masses and dynamics, not for every complex field. An ultralight scalar and a heavy neutral scalar need not resemble each other on galactic scales. A conserved global charge is useful in some complex models, but real scalar models also have viable wave descriptions. The real-versus-complex distinction must not be used to claim all successful scalar phenomenology for only one class.

The observational programme must derive a background history, perturbation transfer function and halo behavior, then confront the relevant data. A simulation such as Schive, Chiueh and Broadhurst (2014) is evidence about a specified model's calculated behavior; it is not itself a measurement of an ultra-faint dwarf. Bounds from the Lyman-alpha forest constrain modeled suppression and its parameters. They do not measure an imaginary axis. The programme gains credibility by keeping the theoretical prediction, the observation and the inference between them separate.

## 5. Complexification and singularities

A complex extension does not generally remove a singularity. The function $1/z$ still has a pole at $z = 0$. A contour can avoid the pole, and an integral around it can have a well-defined residue, without making the function regular at the excluded point. Avoiding a point during a calculation is not equivalent to removing it from the theory's physical geometry.

General-relativistic singularity theorems concern geodesic incompleteness under specified assumptions. They are not all statements that a particular coordinate component or curvature scalar diverges. A proposed resolution must identify the relevant spacetime, geodesics and physical continuation rather than demonstrate only that a complex contour can be drawn.

The Hartle-Hawking no-boundary proposal illustrates a substantive use of complex or Euclidean methods in quantum cosmology (Hartle and Hawking, 1983). It involves a proposed quantum state and boundary prescription. It is not a theorem that every singular Lorentzian solution becomes regular through analytic continuation.

A candidate resolution should establish what replaces the classical description, which observables remain finite or well-defined and whether physically relevant

evolution can be continued. If the construction changes the state, contour, dynamics or admissible metrics, those changes are part of the proposal. They cannot be declared free because they are expressed in complex variables.

Complex structure may participate in a successful resolution. The elementary pole example shows why it does not guarantee one.

### 5.1 A criterion for genuine resolution

Take f(z) = 1/z. On a circle of nonzero radius around the origin, its contour integral is finite, yet f remains undefined at the origin. This separates three properties: avoiding a singular point, assigning an integral a value and extending the underlying object regularly through that point. A gravitational argument needs to identify which property it establishes.

For a proposed complex spacetime, the analogous requirements include a physically interpreted metric or observable replacement, a specification of which histories are admitted, and an account of continuation or completeness appropriate to that theory. A regular Euclidean cap can be part of a quantum-state proposal without providing a Lorentzian extension of every black-hole interior. Nor does geodesic incompleteness necessarily identify a curvature scalar that diverges. The singularity theorems and a proposed quantum resolution must be compared at the level of their precise claims.

Complexification is therefore a tool with potentially important consequences, not a universal removal operation. One successful regular solution would be a genuine result. A proof for a whole class would be stronger. Neither follows from the existence of paths around a pole.

## 6. Real and complex quantum theories

Renou and colleagues studied network scenarios in which real and complex quantum models give different correlations under specified source-independence and composition assumptions (Renou et al., 2021). Chen and colleagues performed a related experimental test (Chen et al., 2022). These results concern competing physical model classes, not the possibility of replacing each complex coordinate by two real coordinates in an equivalent description.

The distinction has become sharper in subsequent work. Hoffreumon and Woods's March 2026 preprint argues that operational independence differs from a product-state requirement on the mathematical source states. Under their operational formulation, they establish a broader reproduction of complex-theory correlations by real models. This is a preprint's stated result, not a declaration that every interpretive question is settled (Hoffreumon and Woods, 2026).

The safe conclusion is substantive: the composition and independence rules matter. An experiment can distinguish theories that share some algebraic ingredients while differing in those rules. An empirically equivalent reformulation cannot be excluded merely because its coordinates are real. Any proposal to

derive quantum theory from a deeper substrate must reproduce the successful correlations and their relevant operational structure.

These findings support careful attention to complex structure without establishing the imaginary-sector dark-matter proposal. The latter requires its own coupling mechanism. The evidence should not be transferred from one claim to the other through the shared word complex.

## 7. Phase, amplitude and measurement

For a complex action S = A + iB, the expression exp(iS/hbar) factors into exp(iA/hbar) times exp(-B/hbar). The real and imaginary parts therefore have distinct roles in this expression: phase and magnitude. That mathematical identity can guide a model, but it does not show that the classical action is always the real projection of an independently specified physical complex action.

Likewise, obtaining a real-valued measurement result does not imply that a detector measures only the real part of a wavefunction. Quantum expectation values of Hermitian observables are real even though they depend on the complex state and its phase relations. A real output and a real-component projection are different operations.

The measurement problem is not resolved by relabeling a record as a projection from the complex world. A proposed account must explain probabilities, records and the emergence or selection of outcomes under its dynamics and interpretation. Decoherence, branching and objective collapse make different claims; complex notation alone chooses none of them.

## 8. A programme with discriminating tests

The four questions now have separate completion conditions. A gravitational unification must define a common structure and recover the relevant dynamical regimes. A dark-sector model must specify its invariant visible-dark distinction and predict gravitational and cosmological observables. A singularity proposal must state what is continued and under which physical criteria. A quantum-formalism proposal must identify its composition and preparation assumptions and the observations they imply.

The programme becomes stronger if one construction meets several of these conditions with shared parameters or symmetries. That would be evidence for unification. At present, placing several established research programmes beside one another does not demonstrate that their successes arise from the same new mechanism.

A useful first model need not solve all four problems. It could define a complex field with a physically meaningful coupling distinction and test its transformation properties. It could compare two fully specified quantum network models.

It could establish regularity for one class of solutions. Each would contribute more than a broad claim of resolution without the connecting calculation.

The negative checks are equally useful. A gauge-dependent darkness criterion fails. A complex pole remains a pole. A real-coordinate rewriting that preserves all operational predictions is not experimentally rejected by its notation. These checks locate the point where interpretation must become physical construction.

## 9. A completion checklist for the unified proposal

The neutral-scalar example answers the first constructive question: there is a consistent action exhibiting electromagnetic darkness and gravitational sourcing, but it does not implement the literal imaginary-sector thesis. The split-mass example answers a second: a component distinction can become physical when an interaction defines it. These examples narrow the research problem rather than solve the four foundational questions together.

A proposed unifying extension should now give its field content and symmetry group; identify any gauge-invariant sector separation; calculate the constrained spectrum and stability; derive a gravitational response and cosmological preparation; and demonstrate a relation to the quantum-composition or singularity results it invokes. If the same parameters control more than one successful prediction, the unification acquires explanatory force. If each question requires unrelated new assumptions, complex notation alone has not supplied the promised economy.

The target remains positive: find one structure doing several jobs. The present examples show where that structure must be specified and where established mathematics already prevents a shortcut. Mathematical monism is compatible with the programme; empirical support must attach to the particular model that carries it out.

## 10. Conclusion

Complex numbers can describe physical relations efficiently. Their real and imaginary components do not automatically divide nature into visible matter and a hidden gravitational sector.

The calculations identify what a proposal needs: a physical reason for distinguishing components, equations governing them, and resulting observations. The same standard applies to claims about singularities and unification. Mathematical monism motivates a common account; the particular construction must show what it explains.

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